

Contrast Stretching (Numerical)

7	12	8
16	9	6
10	15	1

Contrast Stretching (Numerical)

7	12	8
16	9	6
10	15	1

← Original Image

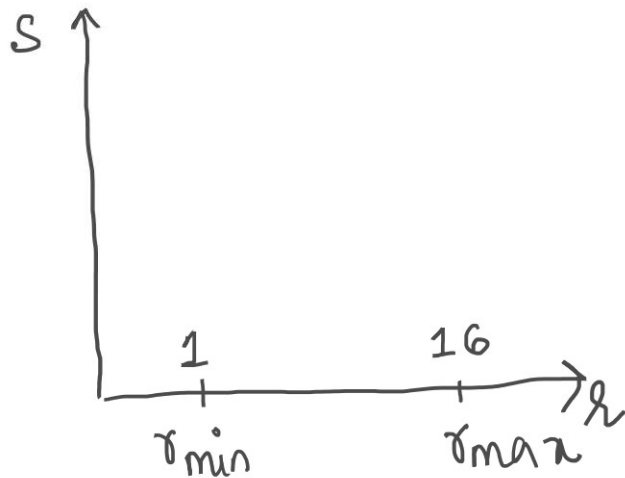
← Perform Contrast Stretching on this image such that the lowest gray level is mapped to 0 and highest to 255

Contrast Stretching (Numerical)

7	12	8
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10	15	1

← Original Image

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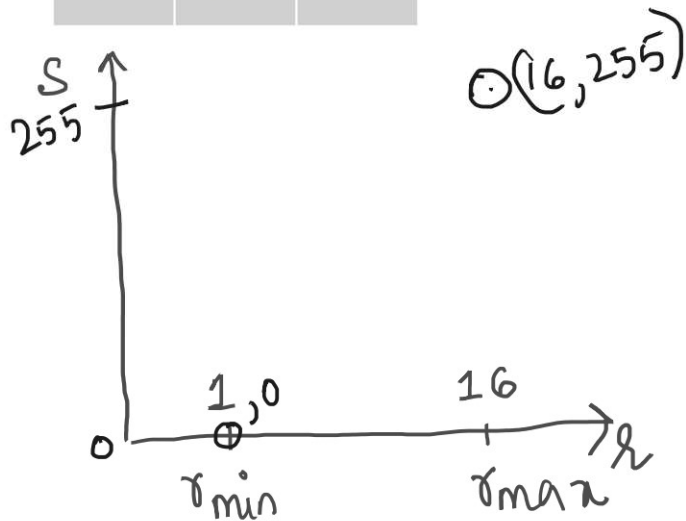


Contrast Stretching (Numerical)

7	12	8
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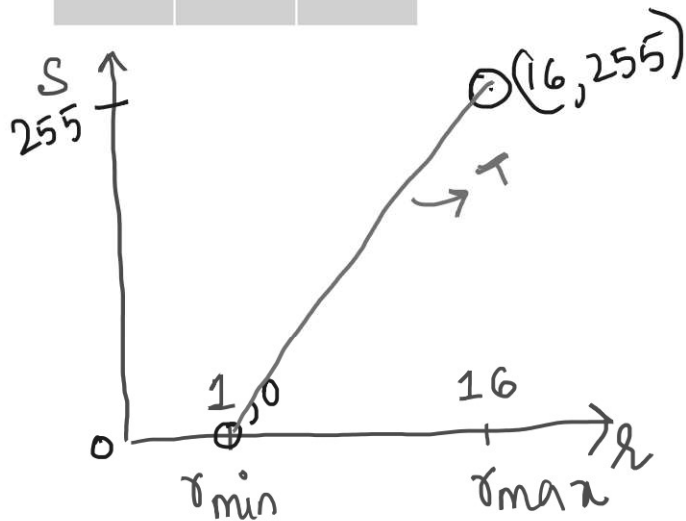


Contrast Stretching (Numerical)

7	12	8
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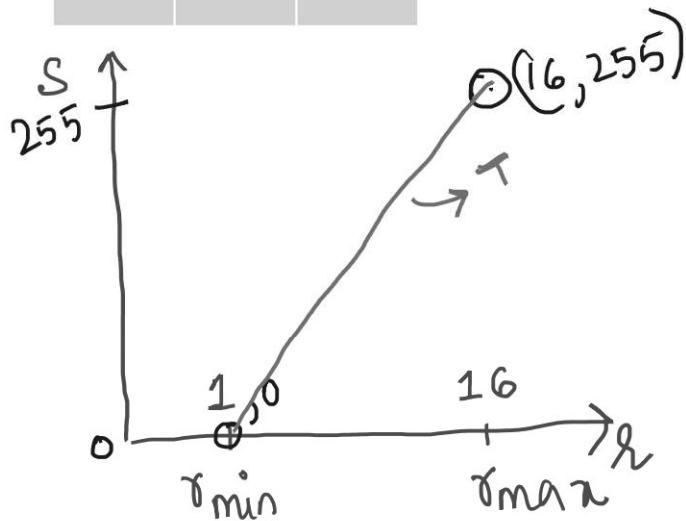


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$$S = mr + c$$

↓ slope

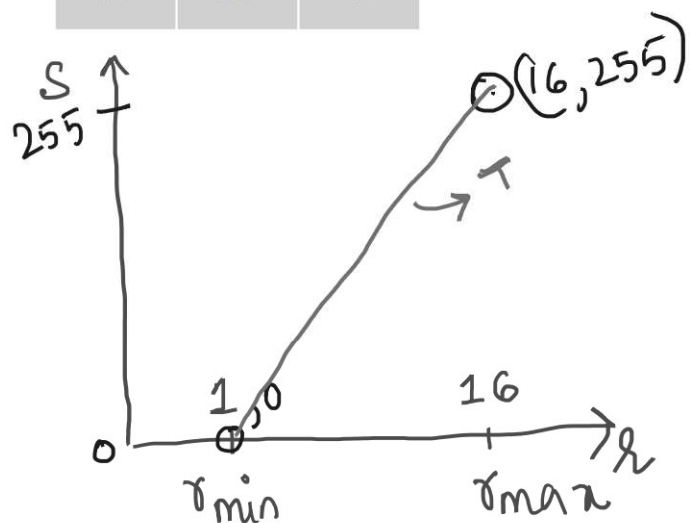
— intercept

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$$S = m r + C \quad \text{— intercept}$$

$$\begin{aligned} \text{slope of line} &= \frac{S_{\max} - S_{\min}}{r_{\max} - r_{\min}} \\ &= \frac{255 - 0}{16 - 1} = \frac{255}{15} = 17 \end{aligned}$$

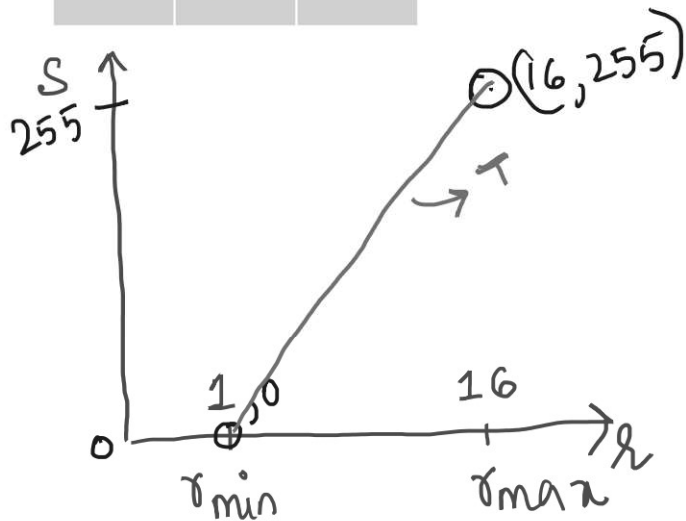
$$\therefore S(x, y) = \left(\frac{S_{\max} - S_{\min}}{r_{\max} - r_{\min}} \right) \times (r - r_{\min}) + S_{\min}$$

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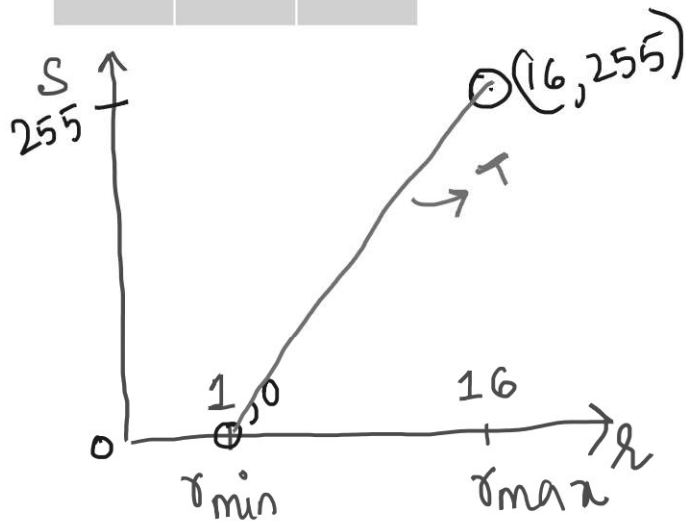
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$$S = m \times (I - I_{\min}) + S_{\min}$$



7
12
8
16
9
6
10
15
1

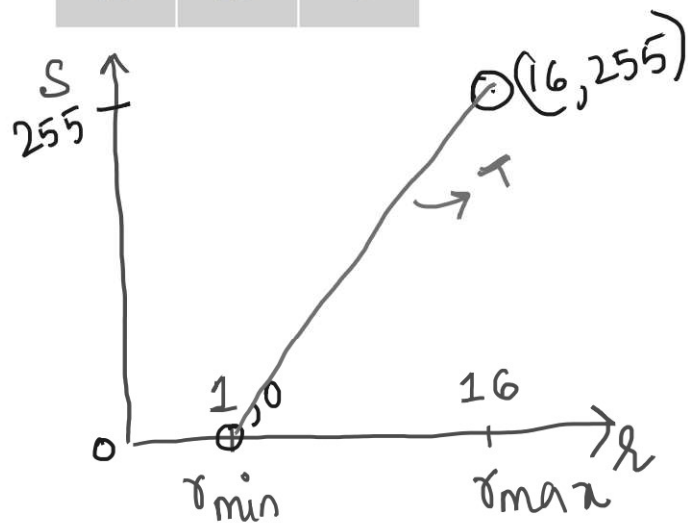
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← Original Image

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$$S = m \times (r - r_{\min}) + S_{\min}$$



7
12
8
16
9
6
10
15
1

$$S = 17 \times (7 - 1) + 0 = 102$$

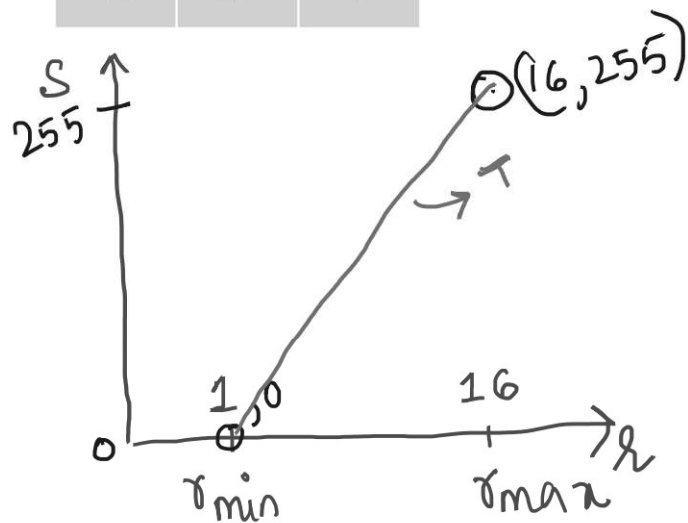
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$$S = m \times (r - r_{\min}) + S_{\min}$$



7
12
8
16
9
6
10
15
1

$$S = 17 \times (7 - 1) + 0 = 102$$

$$S = 17 \times (12 - 1) + 0 = 187$$

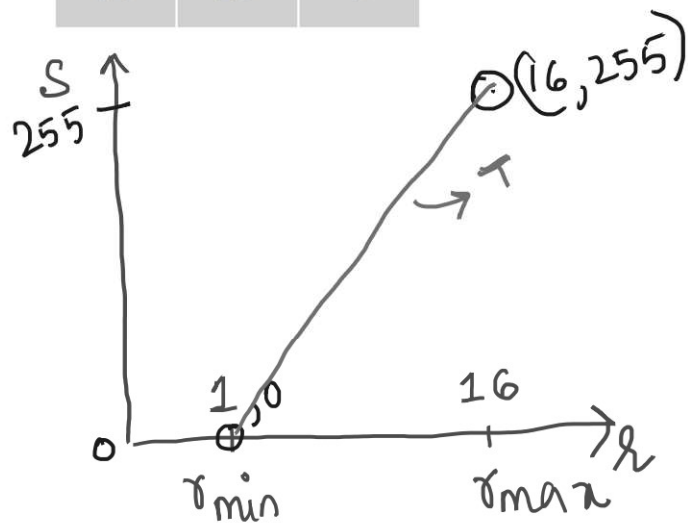
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$$S = m \times (r - r_{\min}) + S_{\min}$$



r	S
7	$17 \times (7 - 1) + 0 = 102$
12	$17 \times (12 - 1) + 0 = 187$
8	$17 \times 7 = 119$
16	$17 \times 15 = 255$
9	$17 \times 8 = 136$
6	$17 \times 5 = 85$
10	$17 \times 9 = 153$
15	$17 \times 14 = 238$
1	$17 \times 0 = 0$

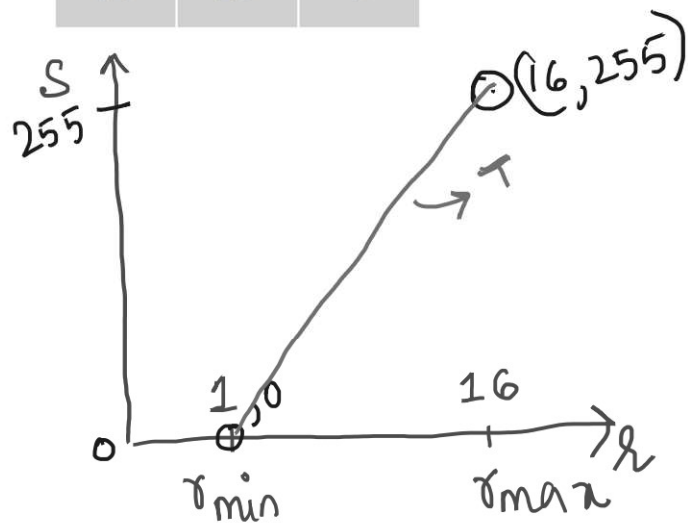
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$$S = m \times (r - r_{\min}) + S_{\min}$$



r	S
7	$17 \times (7 - 1) + 0 = 102$
12	$17 \times (12 - 1) + 0 = 187$
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7	12	8
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$$S = m \times (x - x_{\min}) + S_{\min}$$

x	S	
7	$17 \times (7 - 1) + 0$	$= 102$
12	$17 \times (12 - 1) + 0$	$= 187$
8	17×7	$= 119$
16	17×15	$= 255$
9	17×8	$= 136$
6	17×5	$= 85$
10	17×9	$= 153$
15	17×14	$= 238$
1	17×0	$= 0$

Contrast Stretching (Numerical)

7	12	8
16	9	6
10	15	1

↓
New image

← Original Image

← Perform Contrast Stretching on this image such that the lowest gray level is mapped to 0 and highest to 255

$$S = m \times (r - r_{\min}) + S_{\min}$$

7	S	$17 \times (7 - 1) + 0$	$= 102$
12		$17 \times (12 - 1) + 0$	$= 187$
8		17×7	$= 119$
16		17×15	$= 255$
9		17×8	$= 136$
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Contrast Stretching (Numerical)

7	12	8
16	9	6
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↓
New image

102	187	119
255	136	85
153	238	0

← Original Image

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$$S = m \times (r - r_{\min}) + S_{\min}$$

7	S	$17 \times (7 - 1) + 0$	$= 102$
7		$17 \times (12 - 1) + 0$	$= 187$
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1			

Contrast Stretching (Numerical)

7	12	8
16	9	6
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↓
New image

102	187	119
255	136	85
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1			

Piecewise Linear Transformation or CS (Numerical)

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4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

Piecewise Linear Transformation or CS (Numerical)

4	3	5	2
3	6	4	6
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7	6	4	1

← Original Image

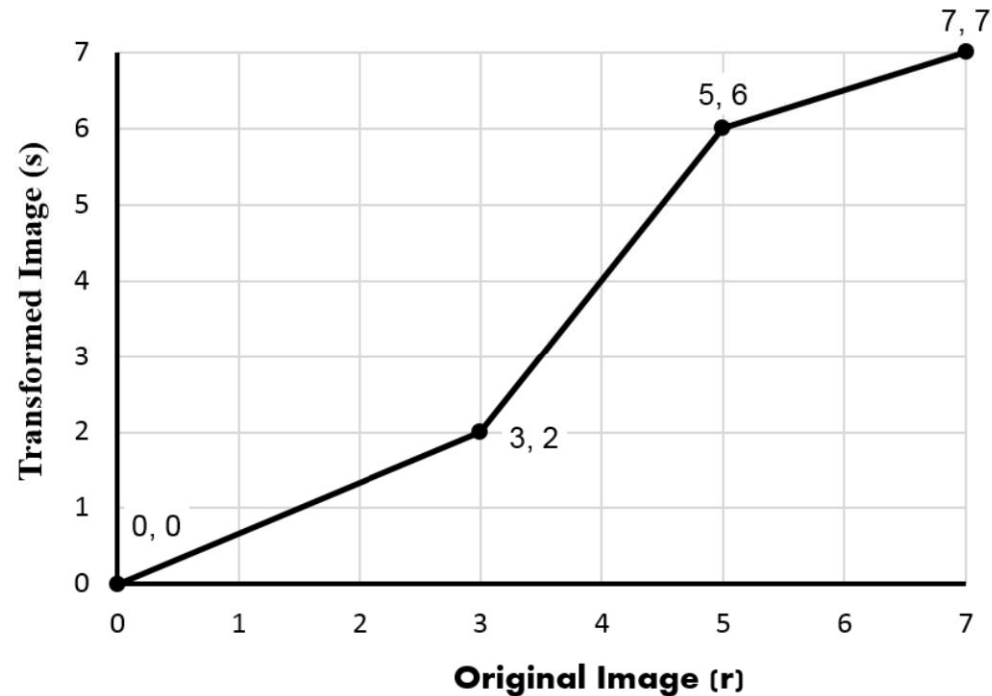
← Perform piecewise Contrast stretching according to graph given

Piecewise Linear Transformation or CS (Numerical)

4	3	5	2
3	6	4	6
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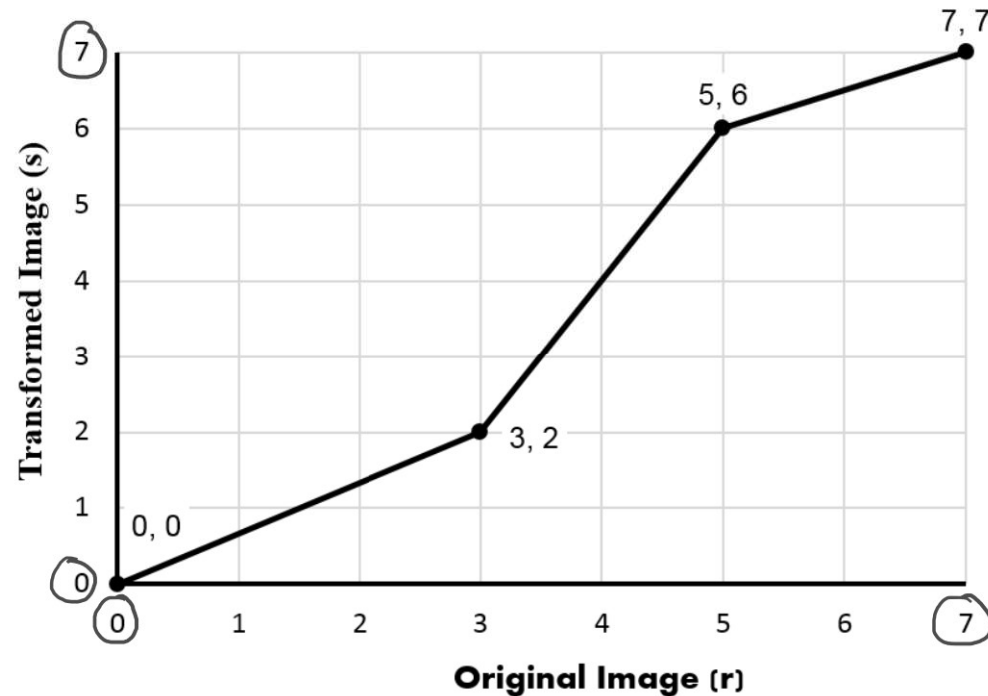


Piecewise Linear Transformation or CS (Numerical)

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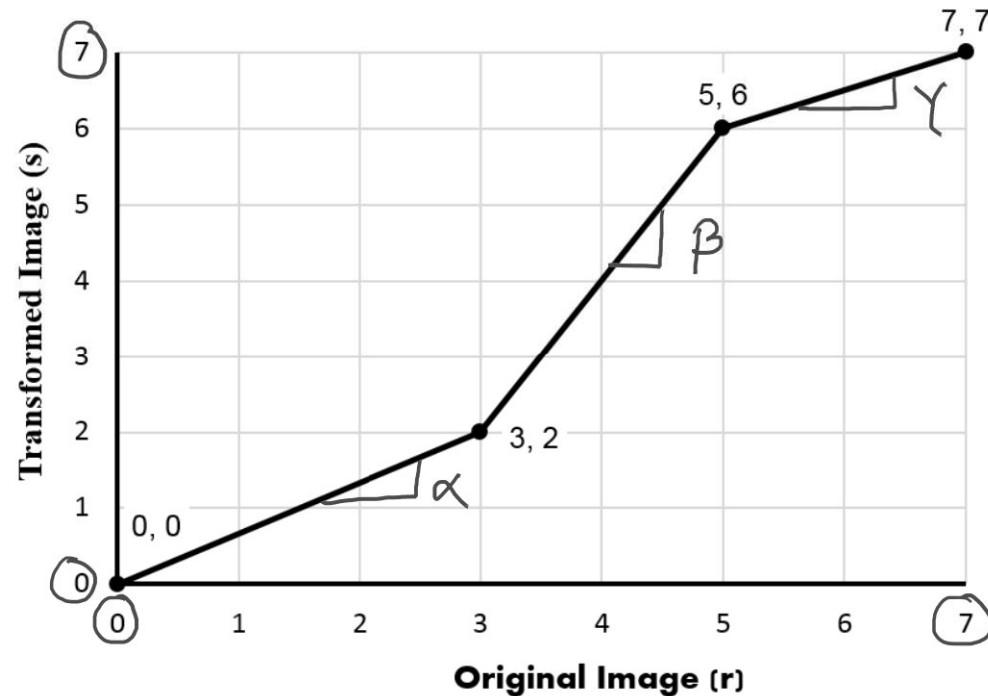


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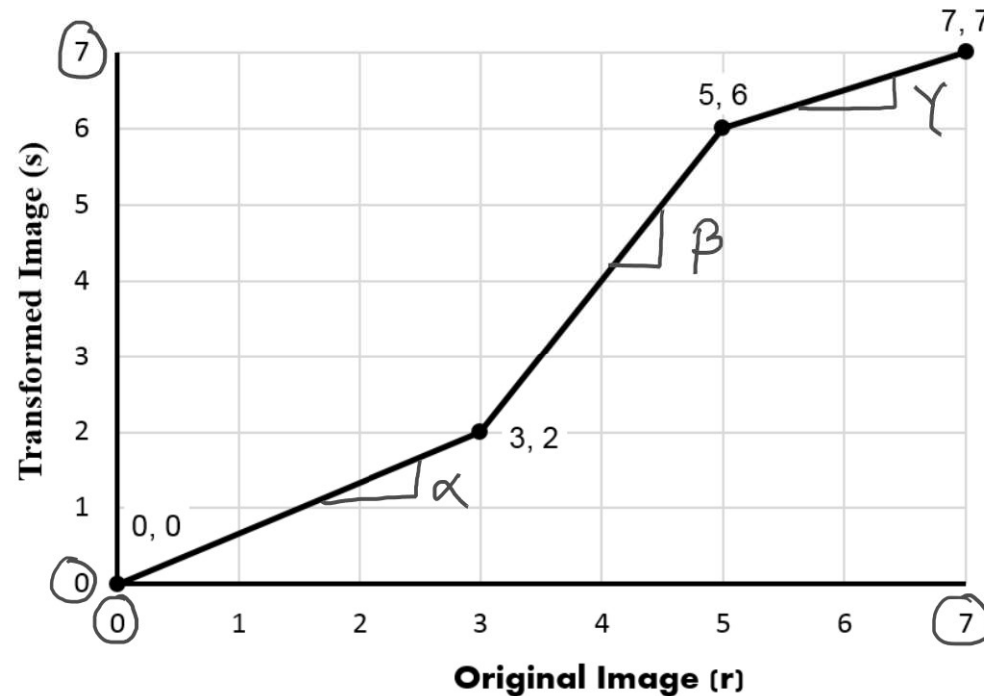
Piecewise Linear Transformation or CS (Numerical)

$$\begin{aligned} r_{\min} &= 0 & s_{\min} &= 0 \\ r_1 &= 3 & s_1 &= 2 \\ r_2 &= 5 & s_2 &= 6 \\ r_{\max} &= 7 & s_{\max} &= 7 \end{aligned}$$

4	3	5	2
3	6	4	6
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7	6	4	1

← Original Image

← Perform piecewise Contrast stretching according to graph given



$$S = \begin{cases} \alpha \cdot r & 0 \leq r < r_1 \end{cases}$$

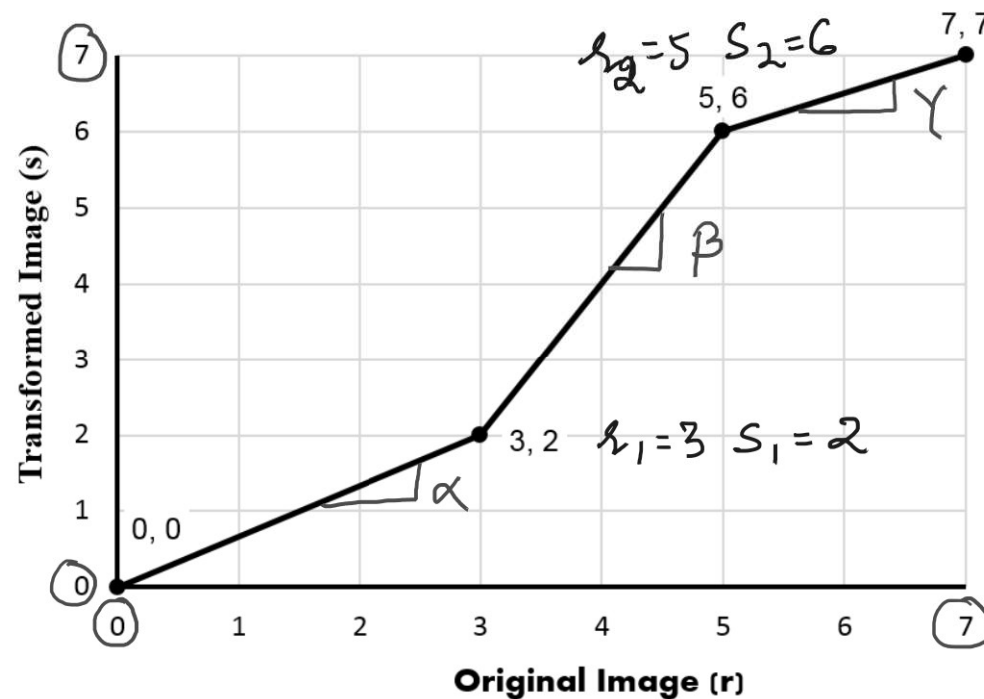
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$$S = \begin{cases} \alpha \cdot r & 0 \leq r < r_1 \end{cases}$$

Piecewise Linear Transformation or CS (Numerical)

$$r_{\min}=0 \quad S_{\min}=0$$

$$r_1=3 \quad S_1=2$$

$$r_2=5 \quad S_2=6$$

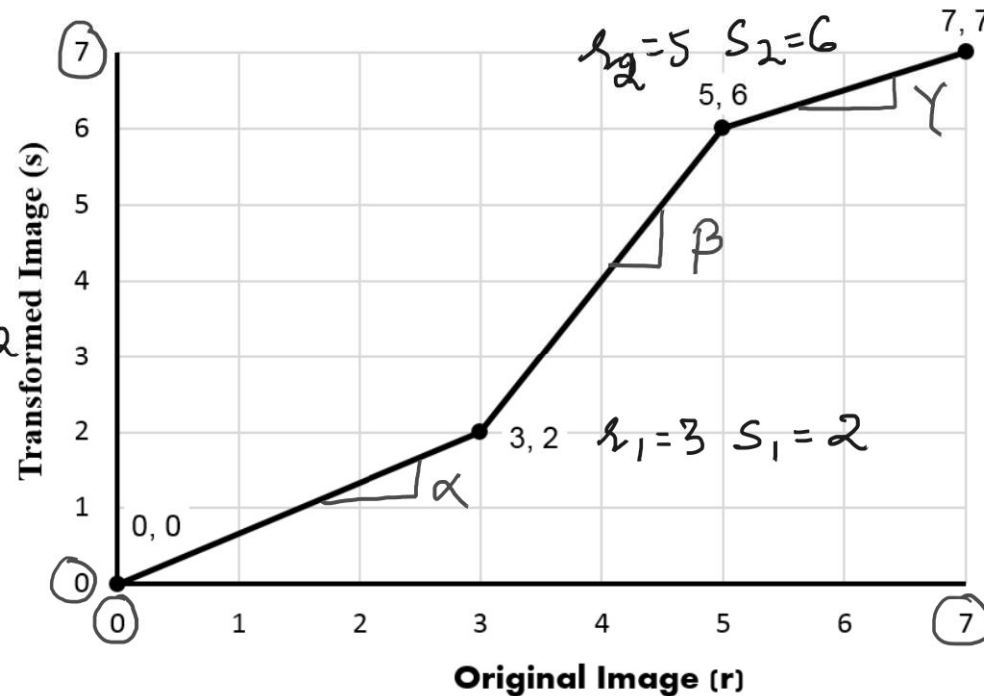
$$r_{\max}=7 \quad S_{\max}=7$$

4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

← Original Image

← Perform piecewise Contrast stretching according to graph given

$$S = \begin{cases} \alpha \cdot r & 0 \leq r < r_1 \\ \beta \cdot (r - r_1) + S_1 & r_1 \leq r < r_2 \\ \gamma \cdot (r - r_2) + S_2 & r_2 \leq r \leq r_{\max} \end{cases}$$



Piecewise Linear Transformation or CS (Numerical)

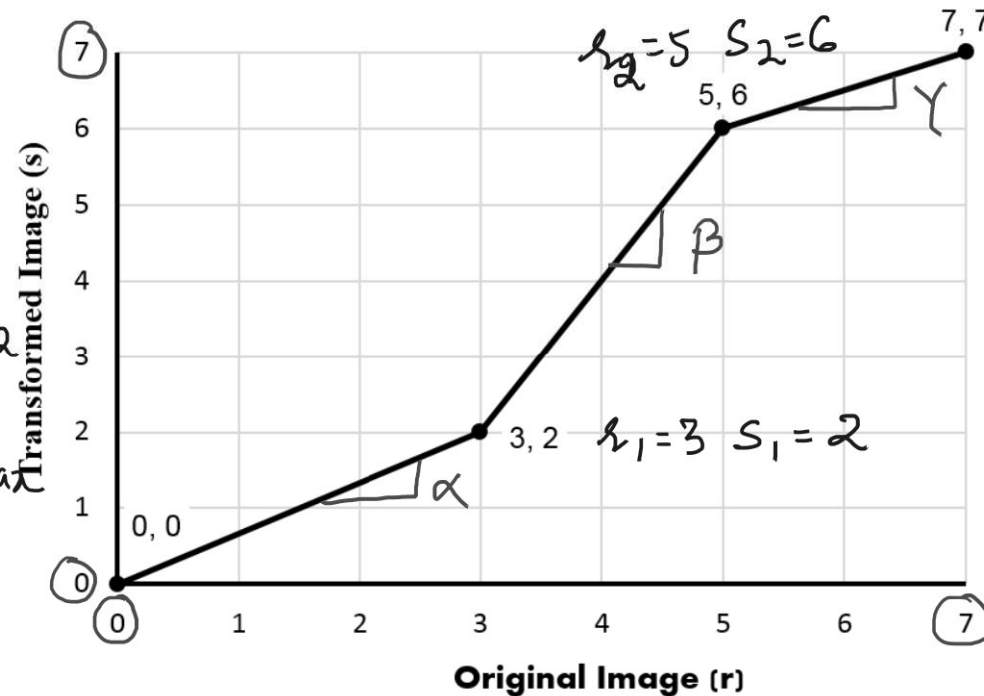
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$$S = \begin{cases} \alpha \cdot r & 0 \leq r < r_1 \\ \beta \cdot (r - r_1) + s_1 & r_1 \leq r < r_2 \\ \gamma \cdot (r - r_2) + s_2 & r_2 \leq r < r_{\max} \end{cases}$$



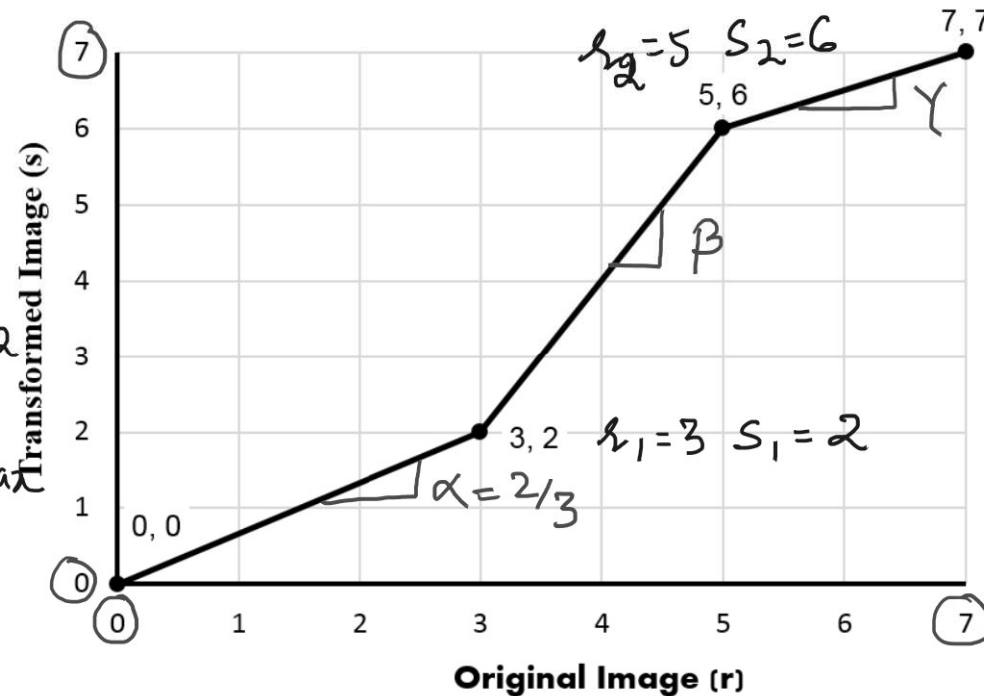
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$$\alpha = \frac{2 - 0}{3 - 0} = 2/3$$

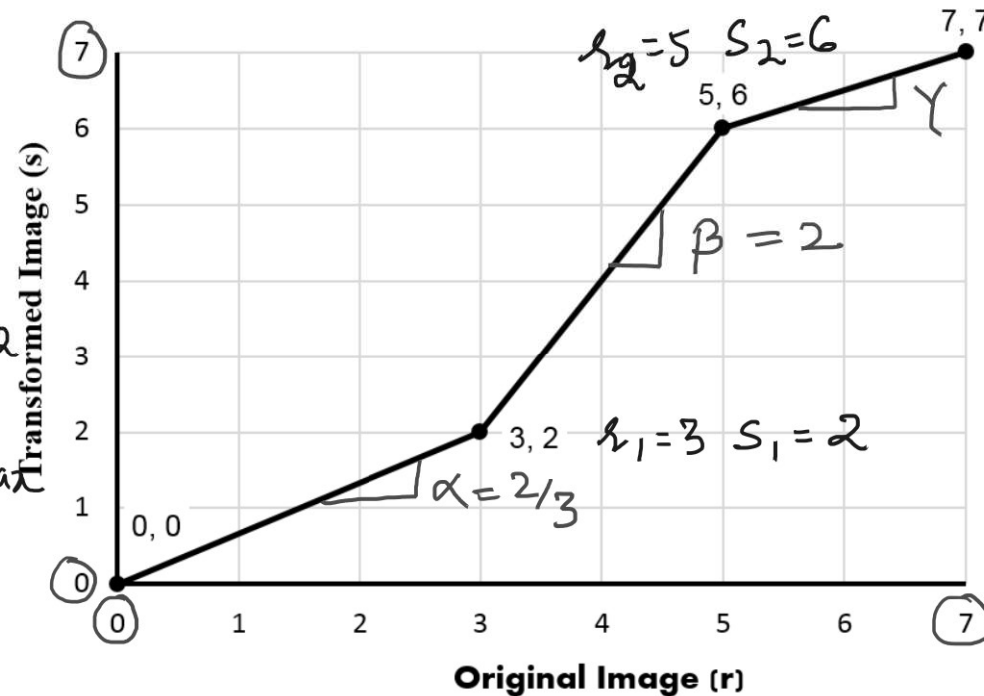
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4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

← Original Image

← Perform piecewise Contrast stretching according to graph given



$$r_{\max}, S_{\max}$$

$$\beta = \frac{6 - 2}{5 - 3} = \frac{4}{2} = 2$$

$$S = \begin{cases} \alpha \cdot r & 0 \leq r < r_1 \\ \beta \cdot (r - r_1) + S_1 & r_1 \leq r < r_2 \\ \gamma \cdot (r - r_2) + S_2 & r_2 \leq r < r_{\max} \end{cases}$$

$$\alpha = \frac{2 - 0}{3 - 0} = \frac{2}{3}$$

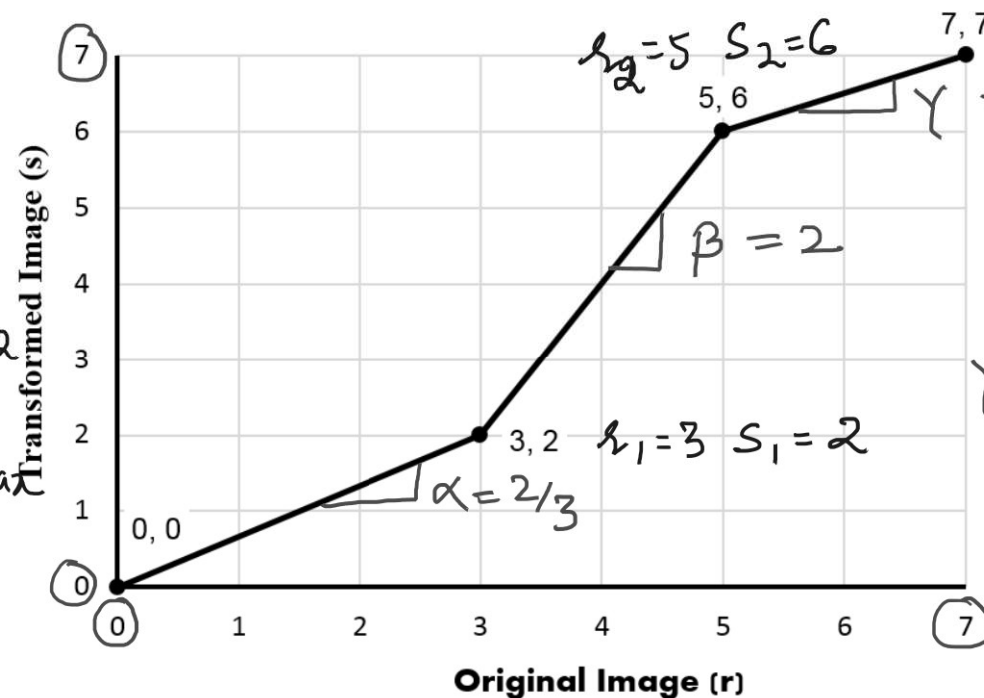
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$$S = \begin{cases} \alpha \cdot r & 0 \leq r < r_1 \\ \beta \cdot (r - r_1) + S_1 & r_1 \leq r < r_2 \\ \gamma \cdot (r - r_2) + S_2 & r_2 \leq r < r_{\max} \end{cases}$$

$$\alpha = \frac{2-0}{3-0} = \frac{2}{3}$$

$$\begin{aligned} \beta &= \frac{6-2}{5-3} = \frac{4}{2} = 2 \\ \gamma &= \frac{7-6}{7-5} = \frac{1}{2} = 0.5 \end{aligned}$$

$$\alpha = 2/3$$

$$\beta = 2$$

$$\gamma = 0.5$$

$$\alpha = 2/3$$

$$\beta = 2$$

$$\gamma = 0.5$$

$$S = \begin{cases} \alpha \cdot \gamma & 0 \leq r < r_1 \\ \beta \cdot (\gamma - \gamma_1) + S_1 & r_1 \leq r < r_2 \\ \gamma \cdot (\gamma - \gamma_2) + S_2 & r_2 \leq r < r_{\max} \end{cases}$$

$$\alpha = 2/3$$

$$\beta = 2$$

$$\gamma = 0.5$$

$$S = \begin{cases} \alpha \cdot x & 0 \leq x < x_1 \\ \beta \cdot (x - x_1) + S_1 & x_1 \leq x < x_2 \\ \gamma \cdot (x - x_2) + S_2 & x_2 \leq x < x_{\max} \end{cases}$$

$$x_{\min} = 0 \quad S_{\min} = 0$$

$$x_1 = 3 \quad S_1 = 2$$

$$x_2 = 5 \quad S_2 = 6$$

$$x_{\max} = 7 \quad S_{\max} = 7$$

$$\alpha = 2/3$$

$$\beta = 2$$

$$\gamma = 0.5$$

$$S = \begin{cases} \alpha \cdot x & 0 \leq x < x_1 \\ \beta \cdot (x - x_1) + S_1 & x_1 \leq x < x_2 \\ \gamma \cdot (x - x_2) + S_2 & x_2 \leq x < x_{\max} \end{cases}$$

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$$x_1 = 3 \quad S_1 = 2$$

$$x_2 = 5 \quad S_2 = 6$$

$$x_{\max} = 7 \quad S_{\max} = 7$$

$$\alpha = 2/3$$

$$\beta = 2$$

$$\gamma = 0.5$$

x

S

0

1

2

3

4

5

6

7

$$S = \begin{cases} \alpha \cdot x & 0 \leq x < x_1 \\ \beta \cdot (x - x_1) + S_1 & x_1 \leq x < x_2 \\ \gamma \cdot (x - x_2) + S_2 & x_2 \leq x < x_{\max} \end{cases}$$

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$$x_1 = 3 \quad S_1 = 2$$

$$x_2 = 5 \quad S_2 = 6$$

$$x_{\max} = 7 \quad S_{\max} = 7$$

$$\alpha = 2/3$$

$$\beta = 2$$

$$\gamma = 0.5$$

x

S

0

$$\alpha \cdot x = 2/3 \times 0 = 0$$

1

2

3

4

5

6

7

$$S = \begin{cases} \alpha \cdot x & 0 \leq x < x_1 \\ \beta \cdot (x - x_1) + S_1 & x_1 \leq x < x_2 \\ \gamma \cdot (x - x_2) + S_2 & x_2 \leq x < x_{\max} \end{cases}$$

$$x_{\min} = 0 \quad S_{\min} = 0$$

$$x_1 = 3 \quad S_1 = 2$$

$$x_2 = 5 \quad S_2 = 6$$

$$x_{\max} = 7 \quad S_{\max} = 7$$

$$\alpha = 2/3$$

$$\beta = 2$$

$$\gamma = 0.5$$

x

S

0

$$\alpha \cdot x = 2/3 \times 0 = 0$$

$$[\alpha \cdot (x - x_{\min}) + S_{\min}]$$

$$S = \begin{cases} \alpha \cdot x & 0 \leq x < x_1 \\ \beta \cdot (x - x_1) + S_1 & x_1 \leq x < x_2 \\ \gamma \cdot (x - x_2) + S_2 & x_2 \leq x < x_{\max} \end{cases}$$

1

2

3

4

5

6

7

$$x_{\min} = 0 \quad S_{\min} = 0$$

$$x_1 = 3 \quad S_1 = 2$$

$$x_2 = 5 \quad S_2 = 6$$

$$x_{\max} = 7 \quad S_{\max} = 7$$

$$\alpha = 2/3$$

$$\beta = 2$$

$$\gamma = 0.5$$

x

S

0

$$\alpha \cdot x = 2/3 \times 0 = 0 \quad [\alpha \cdot (x - x_{\min}) + S_{\min}]$$

1

$$\alpha \cdot x = 2/3 \times 1 = 0.67 \approx 1$$

2

$$\alpha \cdot x = 2/3 \times 2 = 1.33 \approx 1$$

3

4

5

6

7

$$S = \begin{cases} \alpha \cdot x & 0 \leq x < x_1 \\ \beta \cdot (x - x_1) + S_1 & x_1 \leq x < x_2 \\ \gamma \cdot (x - x_2) + S_2 & x_2 \leq x < x_{\max} \end{cases}$$

$$x_{\min} = 0 \quad S_{\min} = 0$$

$$x_1 = 3 \quad S_1 = 2$$

$$x_2 = 5 \quad S_2 = 6$$

$$x_{\max} = 7 \quad S_{\max} = 7$$

$$\alpha = 2/3 \quad \beta = 2 \quad \gamma = 0.5$$

x	S
0	$\alpha \cdot x = 2/3 \times 0 = 0$
1	$\alpha \cdot x = 2/3 \times 1 = 0.67 \approx 1$
2	$\alpha \cdot x = 2/3 \times 2 = 1.33 \approx 1$
3	$\beta \cdot (x - x_1) + S_1 = 2 \times (3 - 3) + 2 = 2$
4	
5	
6	
7	

$$S = \begin{cases} \alpha \cdot x & 0 \leq x < x_1 \\ \beta \cdot (x - x_1) + S_1 & x_1 \leq x < x_2 \\ \gamma \cdot (x - x_2) + S_2 & x_2 \leq x < x_{\max} \end{cases}$$

$$x_{\min} = 0 \quad S_{\min} = 0$$

$$x_1 = 3 \quad S_1 = 2$$

$$x_2 = 5 \quad S_2 = 6$$

$$x_{\max} = 7 \quad S_{\max} = 7$$

$$\alpha = 2/3 \quad \beta = 2 \quad \gamma = 0.5$$

x	S
0	$\alpha \cdot x = 2/3 \times 0 = 0$
1	$\alpha \cdot x = 2/3 \times 1 = 0.67 \approx 1$
2	$\alpha \cdot x = 2/3 \times 2 = 1.33 \approx 1$
3	$\beta \cdot (x - x_1) + S_1 = 2 \times (3 - 3) + 2 = 2$
4	$\beta \cdot (x - x_1) + S_1 = 2 \times (4 - 3) + 2 = 4$
5	
6	
7	

$$S = \begin{cases} \alpha \cdot x & 0 \leq x < x_1 \\ \beta \cdot (x - x_1) + S_1 & x_1 \leq x < x_2 \\ \gamma \cdot (x - x_2) + S_2 & x_2 \leq x < x_{\max} \end{cases}$$

$$x_{\min} = 0 \quad S_{\min} = 0$$

$$x_1 = 3 \quad S_1 = 2$$

$$x_2 = 5 \quad S_2 = 6$$

$$x_{\max} = 7 \quad S_{\max} = 7$$

$$\alpha = 2/3 \quad \beta = 2 \quad \gamma = 0.5$$

$$r \quad S$$

$$0 \quad \alpha \cdot r = 2/3 \times 0 = 0 \quad [\alpha \cdot (r - r_{\min}) + S_{\min}]$$

$$1 \quad \alpha \cdot r = 2/3 \times 1 = 0.67 \approx 1$$

$$2 \quad \alpha \cdot r = 2/3 \times 2 = 1.33 \approx 1$$

$$3 \quad \beta \cdot (r - r_1) + S_1 = 2 \times (3 - 3) + 2 = 2$$

$$4 \quad \beta \cdot (r - r_1) + S_1 = 2 \times (4 - 3) + 2 = 4$$

$$5 \quad \gamma \cdot (r - r_2) + S_2 = 0.5 \cdot (5 - 5) + 6 = 6$$

6

7

$$S = \begin{cases} \alpha \cdot r & 0 \leq r < r_1 \\ \beta \cdot (r - r_1) + S_1 & r_1 \leq r < r_2 \\ \gamma \cdot (r - r_2) + S_2 & r_2 \leq r < r_{\max} \end{cases}$$

$$r_{\min} = 0 \quad S_{\min} = 0$$

$$r_1 = 3 \quad S_1 = 2$$

$$r_2 = 5 \quad S_2 = 6$$

$$r_{\max} = 7 \quad S_{\max} = 7$$

$$\alpha = 2/3 \quad \beta = 2 \quad \gamma = 0.5$$

$$h \quad S$$

$$0 \quad \alpha \cdot h = 2/3 \times 0 = 0 \quad [\alpha \cdot (h - h_{\min}) + S_{\min}]$$

$$1 \quad \alpha \cdot h = 2/3 \times 1 = 0.67 \approx 1$$

$$2 \quad \alpha \cdot h = 2/3 \times 2 = 1.33 \approx 1$$

$$3 \quad \beta \cdot (h - h_1) + S_1 = 2 \times (3 - 3) + 2 = 2$$

$$4 \quad \beta \cdot (h - h_1) + S_1 = 2 \times (4 - 3) + 2 = 4$$

$$5 \quad \gamma \cdot (h - h_2) + S_2 = 0.5 \cdot (5 - 5) + 6 = 6$$

$$6 \quad \gamma \cdot (h - h_2) + S_2 = 0.5 \cdot (6 - 5) + 6 = 6.5 \approx 7$$

$$7 \quad \gamma \cdot (h - h_2) + S_2 = 0.5 \cdot (7 - 5) + 6 = 7$$

$$S = \begin{cases} \alpha \cdot h & 0 \leq h < h_1 \\ \beta \cdot (h - h_1) + S_1 & h_1 \leq h < h_2 \\ \gamma \cdot (h - h_2) + S_2 & h_2 \leq h < h_{\max} \end{cases}$$

$$h_{\min} = 0 \quad S_{\min} = 0$$

$$h_1 = 3 \quad S_1 = 2$$

$$h_2 = 5 \quad S_2 = 6$$

$$h_{\max} = 7 \quad S_{\max} = 7$$

$$\alpha = 2/3 \quad \beta = 2 \quad \gamma = 0.5$$

x

0

$$S = \alpha \cdot x = 2/3 \times 0 = 0 \quad [\alpha \cdot (x - x_{\min}) + S_{\min}]$$

1

$$\alpha \cdot x = 2/3 \times 1 = 0.67 \approx 1$$

2

$$\alpha \cdot x = 2/3 \times 2 = 1.33 \approx 1$$

3

$$\beta \cdot (x - x_1) + S_1 = 2 \times (3 - 3) + 2 = 2$$

4

$$\beta \cdot (x - x_1) + S_1 = 2 \times (4 - 3) + 2 = 4$$

5

$$\gamma \cdot (x - x_2) + S_2 = 0.5 \cdot (5 - 5) + 6 = 6$$

6

$$\gamma \cdot (x - x_2) + S_2 = 0.5 \cdot (6 - 5) + 6 = 6.5 \approx 7$$

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$$S = \begin{cases} \alpha \cdot x & 0 \leq x \leq x_1 \\ \beta \cdot (x - x_1) + S_1 & x_1 \leq x \leq x_2 \\ \gamma \cdot (x - x_2) + S_2 & x_2 \leq x < x_{\max} \end{cases}$$

x

0

$$\alpha \cdot x = 2/3 \times 0 = \underline{0} \quad [\alpha \cdot (x - x_{\min}) + S_{\min}]$$

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$$\gamma \cdot (x - x_2) + S_2 = 0.5 \cdot (5 - 5) + 6 = \underline{6} \leftarrow 2 \times (5 - 3) + 2 = 6$$

6

$$\gamma \cdot (x - x_2) + S_2 = 0.5 \cdot (6 - 5) + 6 = 6.5 \approx \underline{7}$$

7

$$\gamma \cdot (x - x_2) + S_2 = 0.5 \cdot (7 - 5) + 6 = \underline{7}$$

Piecewise Linear Transformation or CS (Numerical)

Piecewise Linear Transformation or CS (Numerical)

4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

Piecewise Linear Transformation or CS (Numerical)

4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

← Original Image

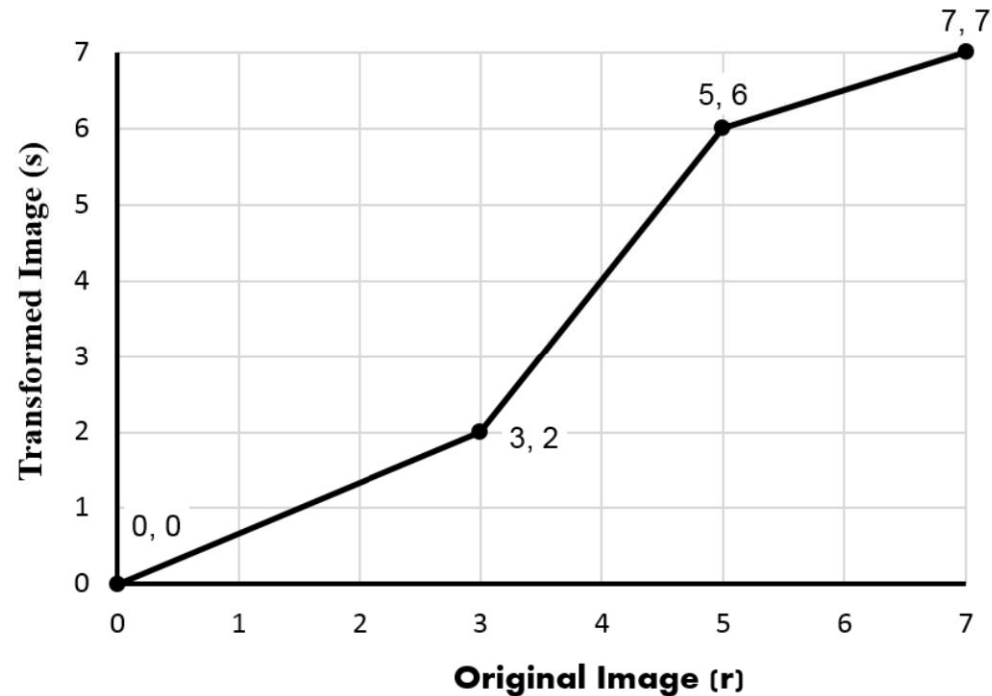
← Perform piecewise Contrast stretching according to graph given

Piecewise Linear Transformation or CS (Numerical)

4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

← Original Image

← Perform piecewise Contrast stretching according to graph given

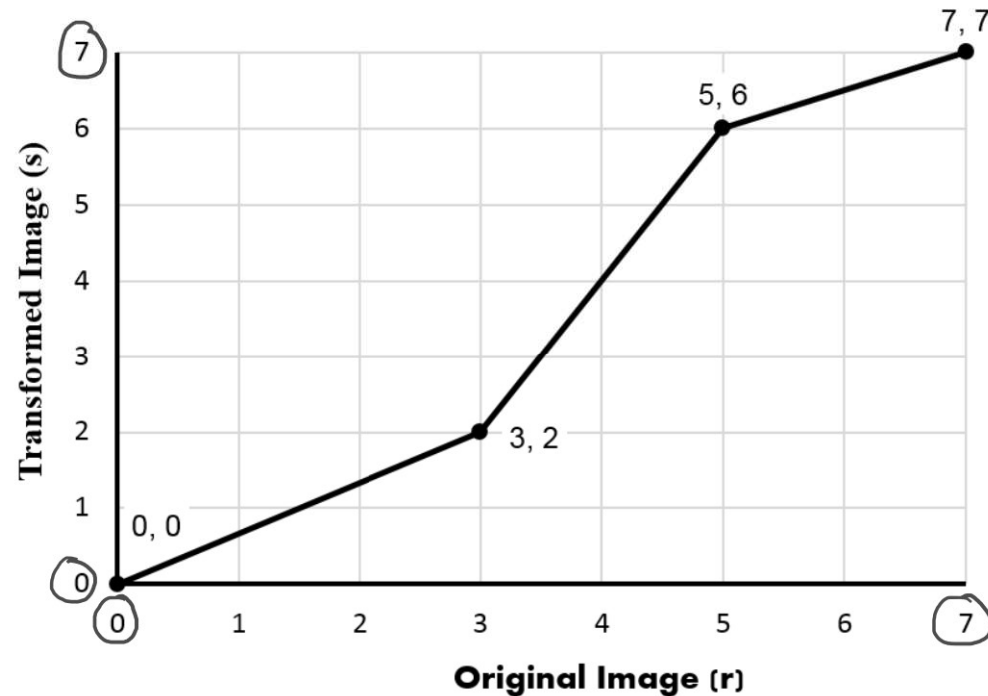


Piecewise Linear Transformation or CS (Numerical)

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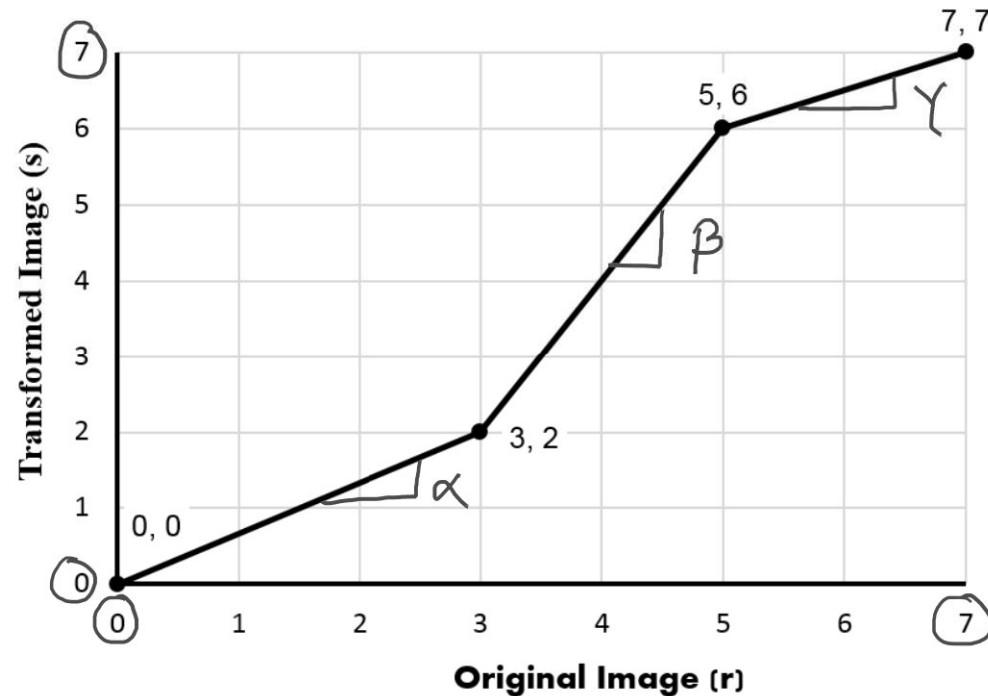


Piecewise Linear Transformation or CS (Numerical)

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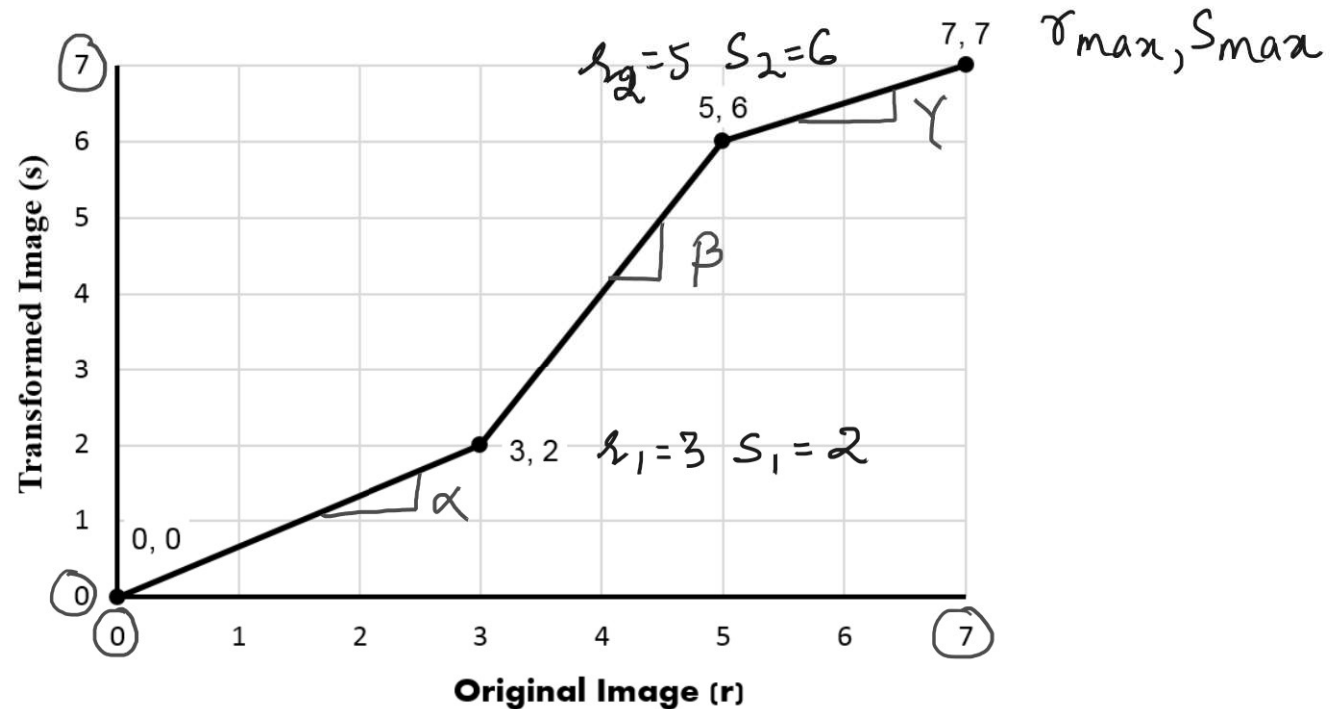


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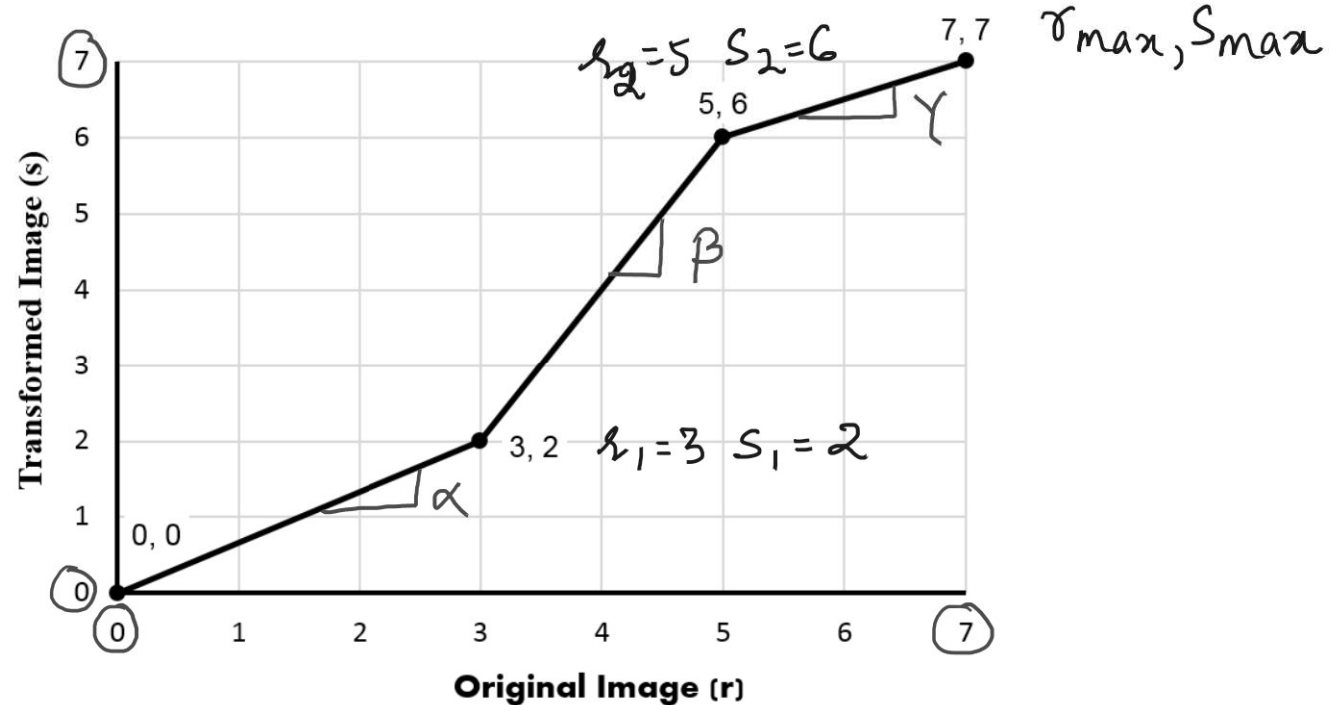


Piecewise Linear Transformation or CS (Numerical)

4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

← Original Image

← Perform piecewise Contrast stretching according to graph given



New Image

Piecewise Linear Transformation or CS (Numerical)

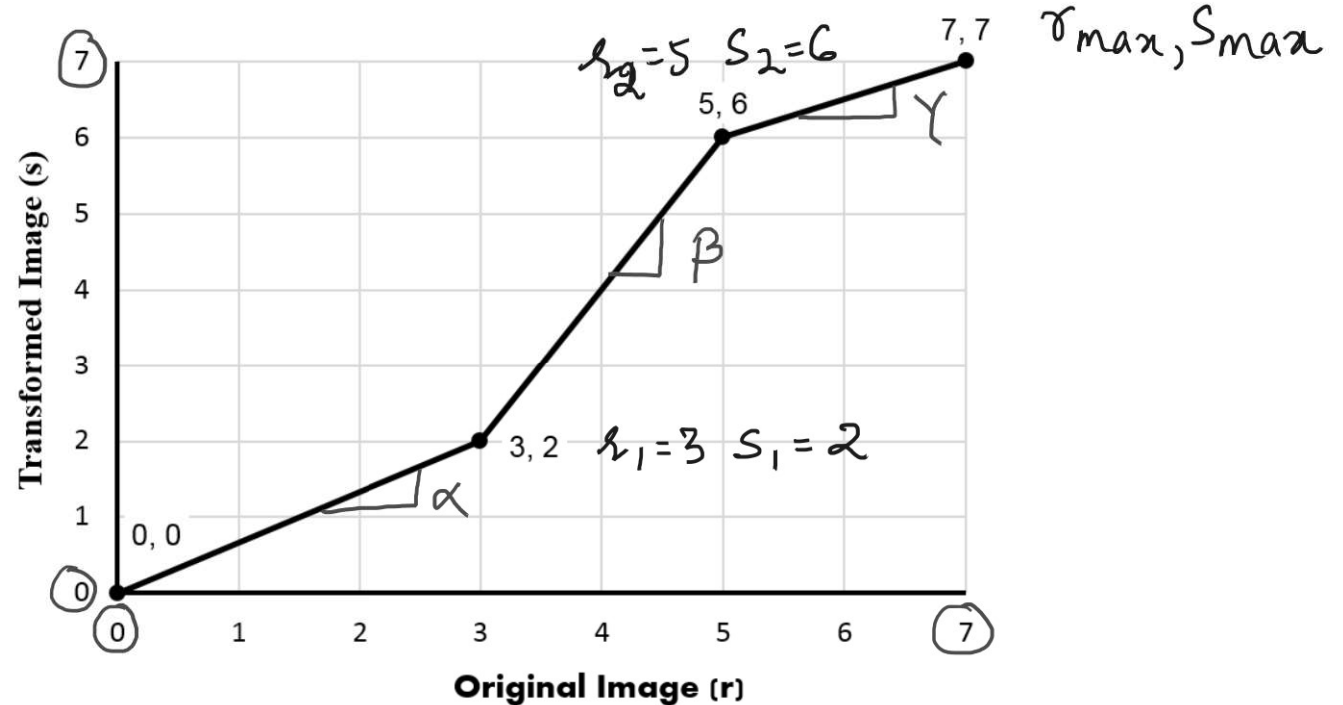
4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

← Original Image

← Perform piecewise Contrast stretching according to graph given

New Image

4	2	6	1
2	7	4	7
1	1	7	6
7	7	4	1



Piecewise Linear Transformation or CS (Numerical)

4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

← Original Image

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New Image

4	2	6	1
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1	1	7	6
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